ISAAC RIVERA TAPIA

Physics MSc candidate working on microscopic optical potentials for deformed nuclei. Seeking a PhD in theoretical nuclear physics starting 2028.

RELEVANT COURSES

- Quantum Collision Theory
- Introduction to Nuclear Physics
- Mechanics of the Continuum
- Effective theories in Physics
- Statistical Mechanics

COMPETENCES

Programming:

- Python
- Fortran
- Matlab
- Linux
- LaTeX

LANGUAGES

Spanish (Native)
English (C1, Oxford Placement Test, 2026)
French (A1, in progress)

CONTACT

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Santiago, Chile

REFERENCES

Dr. Hugo Arellano - U. de Chile
arellano@dfi.uchile.cl
Dr. Nicolás Mujica - U. de Chile
nmujica@dfi.uchile.cl
Dr. Patricio Aceituno - U. de Chile
paceituno2015@gmail.com

AWARDS / EXTRACURRICULAR ACTIVITIES

Gold Medal, SOCHIFI Physics Olympiad 2021.
Best Student of the Graduating class, Colegio Agustiniano (2021).
Participant, Ibero-American Physics Olympiad 2021.

WORK EXPERIENCE

Research Project, Nuclear Reaction Theory | Universidad de Chile | Mar 2025 - present. Supervisor: Dr. Hugo Arellano

- Developing a microscopic optical potential for scattering from deformed nuclei, built on the effective interaction (g-matrix) from Brueckner-Bethe-Goldstone theory.
- Formulated the potential using a multipole expansion of the nuclear density and a functional expansion of the effective interaction.
- Implementing the model numerically (Fortran, Python) to compute differential cross sections for comparison with experimental data.

Research Internship, Granular Matter | Universidad de Chile | Dec 2025 - Jan 2026. Supervisor: Dr. Nicolás Mujica

- Contributed to the design of a shorter vacuum-tower experiment to measure the charge of particles from their trajectories in a uniform electric field, using simulations to show that a fixed camera over a shorter range could replace the falling camera, a major source of error.
- Modeled Lorentzian distributions to quantify measurement uncertainty and evaluate the precision of the proposed setup.

EDUCATION

MSc in Physics (in progress), Universidad de Chile, 2026 - expected 2027
Research: microscopic optical potentials for deformed nuclei.
BSc in Physics, Universidad de Chile, 2022-2026
GPA 6.2/7.0 (Chilean scale, passing 4.0). **Ranked 1st of 9 graduates of the program in 2026** (official certificate available).

PRESENTATIONS

- "Optical Potential for Deformed Nuclei", I. Rivera. Poster presentation, XXV Chilean Physics Symposium (SOCHIFI), Santiago, Chile, 23-26 Nov 2026 (accepted).
- "Optical Potential for Deformed Nuclei", I. Rivera. Poster presentation, École Joliot-Curie "Theory to Shape Experiments", Saint-Pierre d'Oléron, France, 13-18 Sep 2026.
- "Optical Potential for Deformed Nuclei", I. Rivera. Poster presentation, IV Postgraduate Symposium on Engineering, Science and Innovation, Santiago, Chile, 20-21 Aug 2026.

ADDITIONAL TRAINING

- Summer School on Aspects of Classical and Quantum Field Theory (2026), Universidad de Concepción.
- Winter Internship (2024), Photonic Lattices Laboratory.
- 2nd Quantum Computing School (2024), Pontificia Universidad Católica.
- 6th Winter Camp on Competitive Programming (2023), Universidad Técnica Federico Santa María.

TEACHING ASSISTANT

Electromagnetism
with Dr. Simón Riquelme
Universidad de Chile
Aug-Dec 2024

Introduction to Classical Physics
with Dr. Hugo Arellano
Universidad de Chile
Mar-Jul 2025

Experimental Methods
with Dr. Nicolás Mujica
Mechanics
with Dr. Patricio Aceituno
Universidad de Chile
Mar-Jul 2026

Electromagnetism
with Dr. Nicolás Mujica
Universidad de Chile
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